

A seven-dimensional counterexample to derivative realizability

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Status of this manuscript: Independently checked negative resolution (Codex-agent review).

Target: IS-03, [original catalog snapshot](#).

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The original proof draft was generated in a ChatGPT conversation. A separate Codex agent independently checked the complete argument against the exact target; its [detailed review](#) records **PASS**. The mathematical sections below are unchanged from the reviewed draft. This is independent agent verification, not external human peer review or a formal proof certificate. The [submission record](#) preserves the original manuscript and verification history.

The target and the counterexample

The repository asks whether, for every entrywise-nonnegative real matrix A of order $n \geq 5$, the monic polynomial $p'_A(z)/n$ is the characteristic polynomial of an entrywise-nonnegative real matrix of order $n - 1$.¹

Let C_k denote the permutation matrix of a directed cycle of length k , so that $\det(zI - C_k) = z^k - 1$. Set

$$A = \text{diag} \left(\frac{1}{2}, C_2, C_4 \right) = \begin{pmatrix} \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{pmatrix}. \quad (1)$$

This is an entrywise-nonnegative real 7×7 matrix. Its characteristic polynomial is

$$\begin{aligned} p(z) &= (z - \tfrac{1}{2})(z^2 - 1)(z^4 - 1) \\ &= z^7 - \tfrac{1}{2}z^6 - z^5 + \tfrac{1}{2}z^4 - z^3 + \tfrac{1}{2}z^2 + z - \tfrac{1}{2}. \end{aligned} \quad (2)$$

Its normalized derivative is

$$q(z) = \frac{p'(z)}{7} = z^6 - \frac{3}{7}z^5 - \frac{5}{7}z^4 + \frac{2}{7}z^3 - \frac{3}{7}z^2 + \frac{1}{7}z + \frac{1}{7}. \quad (3)$$

Theorem 1. There is no entrywise-nonnegative real 6×6 matrix with characteristic polynomial q . Consequently, the universal assertion in IS-03 is false.

Exact negative-moment certificate

Let $\mu_1, \dots, \mu_6$ be the roots of q , counted with algebraic multiplicity, and put $s_k = \sum_{i=1}^6 \mu_i^k$. Newton's identities determine these numbers without computing the roots.

For a monic degree- d polynomial

$$q(z) = z^d + c_1 z^{d-1} + \dots + c_d,$$

¹Open Problems in Numerical Linear Algebra, IS-03, "Johnson's derivative-realizability conjecture," checked 11 September 2026. [Canonical entry](#).

the identities are

$$\begin{cases} s_k + c_1 s_{k-1} + \dots + c_{k-1} s_1 + k c_k = 0, & 1 \leq k \leq d, \\ s_k + c_1 s_{k-1} + \dots + c_d s_{k-d} = 0, & k > d. \end{cases} \quad (4)$$

Applying (4) to (3) gives the following rational values:

$$\begin{aligned} s_1 &= \frac{3}{7}, & s_2 &= \frac{79}{49}, & s_3 &= \frac{48}{343}, \\ s_4 &= \frac{6731}{2401}, & s_5 &= \frac{5213}{16807}, & s_6 &= \frac{219766}{117649}. \end{aligned} \quad (5)$$

The next identity is particularly short:

$$7s_7 = 3s_6 + 5s_5 - 2s_4 + 3s_3 - s_2 - s_1. \quad (6)$$

After expressing its right-hand side over the denominator $117649 = 7^6$, its numerator is

$$659298 + 182455 - 659638 + 49392 - 189679 - 50421 = -8593.$$

Therefore

$$s_7 = -\frac{8593}{823543} < 0. \quad (7)$$

Proof of Theorem 1. If an entrywise-nonnegative matrix B had characteristic polynomial q , the trace identity for matrix powers, valid without diagonalizability, would give

$$\text{tr}(B^7) = \sum_{i=1}^6 \mu_i^7 = s_7 < 0.$$

But every entry of B^7 is nonnegative, so its trace is nonnegative. This is a contradiction.

Scope and checks

The counterexample satisfies the real field, entrywise nonnegativity, and order condition in the repository. Neither irreducibility nor zero trace is an assumption there. The example has positive trace and is reducible; these are permitted, not exceptions to the target. It does not contradict a result restricted to trace-zero matrices of orders five or six.

The negative moment also precludes a realization after appending any number of zero eigenvalues, since zeros do not change s_7 . Thus no relaxation by zero padding repairs this particular example.

The accompanying verification script checks (2)–(7) with exact rational arithmetic and independently computes the seventh power trace of the companion matrix of q . The companion matrix is used only to check the certificate, not as a proposed nonnegative realization. These diagnostics are separate from the independent proof review linked above.

The repository submission records **Solved**, with a negative-resolution notice and the independent Codex-agent PASS report linked above.

Sources

1. *Open Problems in Numerical Linear Algebra*, IS-03, [canonical statement](#), checked 11 September 2026.
2. S. L. Hoover, D. McCormick, P. Paparella, and A. Thrall, *On the realizability of the critical points of a realizable list*, arXiv:1712.05454, Conjecture 1.2 and §6; *Linear Algebra and its Applications* 555 (2018), 301–313. [Manuscript](#); [publication](#). This is the primary reference identified by the repository for the conjecture, not a source for the counterexample above.